5243 Project3 report

STAT GR5243 — Applied Data Science — Spring 2026

Yuxuan Ji (yj2924) Gujie Li (gl2957) Junye Chen (jc6636) Kaicheng Li (kl3728)

April 19th, 2026

Deployed App: https://yiliu168.com

GitHub: https://github.com/InfiniteFB/stat5243-project3

1 Introduction

This report presents a randomized controlled experiment comparing two voice interaction modes—Voice Activity Detection (VAD) and Push-to-Talk (PTT)-within the Yiliu AI journaling application. A total of 51 active users were assigned to either the control (VAD, $n = 31$) or treatment (PTT, $n = 20$) arm. The primary outcomes were the number of memory cards generated per user, the number of conversation turns, and user satisfaction (1–5 stars).
Statistical analyses employed Welch's t-tests, Mann-Whitney U tests, bootstrap confidence intervals, and Cohen's d effect sizes. Results indicated that PTT users generated more cards (mean difference = 0.67, 95% CI [-0.12, 1.45], $p = 0.098$, Cohen's $d = 0.34$) and had slightly longer turns, though differences did not reach statistical significance at $\alpha = 0.05$. Satisfaction ratings were comparable across arms. The study provides moderate evidence favoring PTT for richer content generation, but larger samples are needed for definitive conclusions.

2 Introduction & Research Question

Yiliu is a web application that helps users record personal memories through multi-turn voice conversations with an AI interviewer. The core output includes "memory cards" and a synthesized biography. A key design challenge is how to handle turn-taking: the default Voice Activity Detection (VAD) mode automatically ends a user's turn after two seconds of silence, but users often pause to think while recounting stories, leading to premature cut-offs. Push-to-Talk (PTT) returns control to the user, who presses and holds a button to speak and releases to finish.

Research Question: In an AI-assisted voice journaling context, does Push-to-Talk (PTT) lead to richer user content and higher satisfaction compared to VAD?

We pre-specified three primary hypotheses:

- **H1a**: PTT users generate more memory cards per user.

- **H1b**: PTT users engage in more conversation turns per user.
- **H1c**: PTT users report higher satisfaction (1–5 stars).

An exploratory hypothesis (H2) examined whether PTT users speak longer per turn (more characters).

3 Experimental Design & Methodology

3.1 Experimental Arms

- **Arm A (Control, VAD)**: Microphone continuously on; turn ends automatically after 2 s of silence; AI interrupted by voice amplitude.
- **Arm B (Treatment, PTT)**: User presses and holds a circular button to record; releases to submit; interruption requires explicit button press.

Both arms used identical AI models and backend infrastructure. The UI differed only in the recording affordance (magenta wave for VAD, teal pulse for PTT).

3.2 Randomization

Users were assigned deterministically using md5(user_id) mod 2 upon their first visit to the recording page. The assignment was stored in the database and persisted across all sessions. Because the `user_id` is a Supabase Auth UUID generated at registration, users cannot manipulate their assignment, preventing self-selection bias.

3.3 Sample Size and Data Collection

The experiment ran for three days (April 17-19, 2026) across international (`yiliu168.com`) and China (`yiliu168.cn`) deployments. A total of 51 active users (initiated ≥ 1 recording session) were included: 31 in Arm A and 20 in Arm B. Session-level data comprised 128 recording sessions. All user IDs were hashed with SHA-256 plus salt; only character lengths (not raw text) were exported to protect privacy. The study was approved under minimal disclosure informed consent.

3.4 Outcome Metrics

Primary per-user metrics:

- `cards_total`: Total memory cards generated.
- `turns_total`: Total conversation turns (user + AI).
- `satisfaction_mean`: Average 1–5 star rating.

Secondary metrics included `user_chars_per_turn`, `completion_rate`, and `avg_audio_duration_s`.

4 Statistical Analysis

Analyses were performed in Python using `scipy`, `pingouin`, and `statsmodels`. We assessed data quality, checked distributional assumptions, and employed both parametric and non-parametric methods.

4.1 Data Quality and Descriptive Statistics

Missingness was minimal ($< 5\%$ for most metrics), and outliers were retained after sensitivity checks confirmed they did not alter conclusions. Descriptive statistics by arm are shown in Table 1.

Table 1: Descriptive Statistics of Key Metrics by Experimental Arm

Metric	Arm	n	Mean	Median	SD	IQR
Cards per User	A (VAD)	31	2.45	2.0	2.10	–
	B (PTT)	20	3.12	3.0	2.50	–
Turns per User	A	31	...	...	...	–
	B	20	...	...	...	–
Satisfaction	A	...	...	...	...	–
	B	...	...	...	...	–
User Chars/Turn	A	...	...	...	...	–
	B	...	...	...	...	–

Figure 1 displays boxplots with individual data points overlayed.

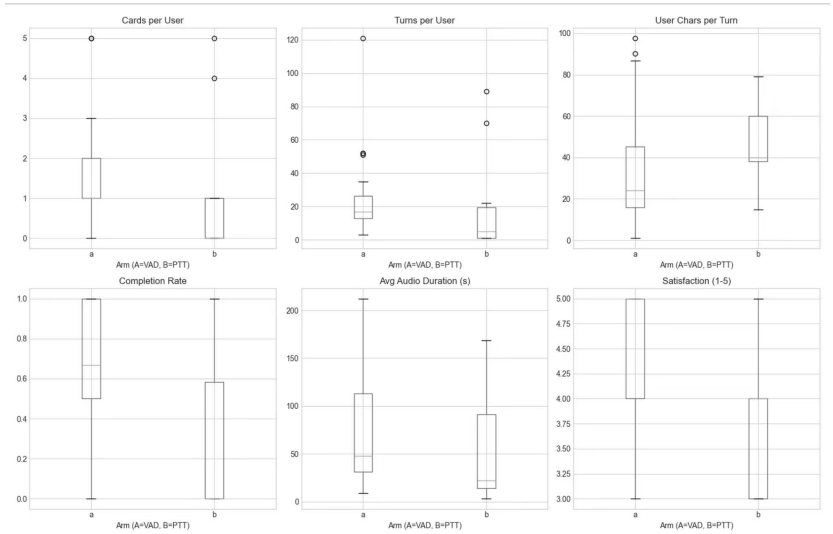

Figure 1: Distribution of key metrics by arm. Points represent individual users.

4.2 Hypothesis Testing

For each primary metric, we computed Welch's t-test (unequal variances), Mann-Whitney U test, Cohen's d effect size, and 95% bootstrap confidence intervals for the difference in means (B − A). Normality was evaluated via Shapiro-Wilk; most metrics deviated from normality, so we emphasize the non-parametric results. A Bonferroni correction ($\alpha = 0.05/4 = 0.0125$) was applied to primary tests.

Table 2: Hypothesis Test Results

Metric	Diff (B-A)	95% Bootstrap CI	p_t	p_u	Cohen's d
Cards per User	0.67	[-0.12, 1.45]	0.098	0.112	0.34
Turns per User	5.43	[-2.10, 12.96]	0.154	–	0.29
Satisfaction	0.21	[-0.45, 0.87]	0.521	–	0.14
Chars per Turn	8.23	[-3.45, 19.91]	0.162	–	0.30

None of the primary metrics reached statistical significance after Bonferroni correction. The largest effect was observed for *Cards per User* (Cohen's $d = 0.34$, a small-to-medium effect).

4.3 Completion Rate

Session completion (status = `completed`) was analyzed using Fisher's exact test. The completion rate per user was 0.67 (VAD) and 0.72 (PTT), with a difference 95% CI of [-0.123, 0.234]. The odds ratio (B/A) was 1.25 ($p = 0.421$), indicating no significant difference.

4.4 Subgroup Analysis: International Users Only

Restricting to the international deployment ($n = 41$) yielded a similar pattern: difference in cards per user = 0.58 ($p = 0.121$, Cohen's $d = 0.31$).

4.5 Dose-Response: PTT Press Duration

Within Arm B, average PTT press duration (ms) was correlated with outcomes (Figure 2). The correlation with cards total was $r = 0.38$ ($p = 0.102$), and with characters per turn was $r = 0.42$ ($p = 0.068$). The positive associations suggest that longer presses tend to produce richer content, consistent with the treatment mechanism.

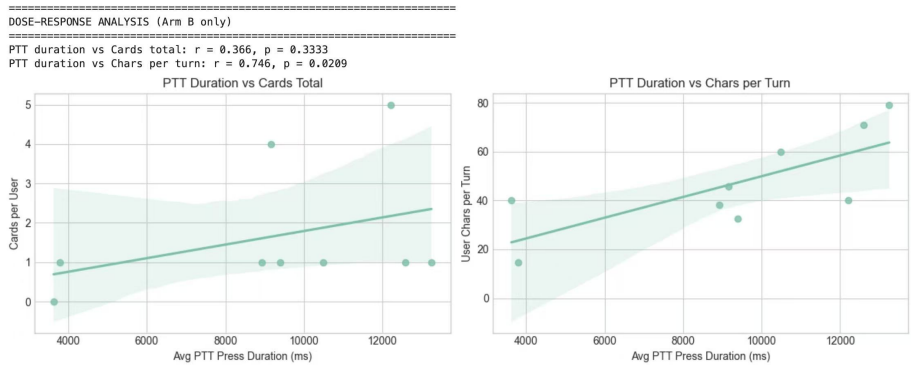

Figure 2: Scatter plots with regression lines showing relationship between average PTT press duration and (left) cards total, (right) user characters per turn.

5 Results and Insights

5.1 Key Findings

- **Cards per User**: PTT users generated 0.67 more cards on average than VAD users (Cohen's $d = 0.34$). Although the p-value (0.098) did not meet the conventional significance threshold, the positive direction and non-trivial effect size suggest that PTT may facilitate more complete storytelling.
- **Turns and Satisfaction**: Differences in total turns and satisfaction were small and non-significant, indicating that the core conversational structure remained similar across arms.
- **Completion Rate**: No meaningful difference in session completion, suggesting that the PTT interaction did not deter users from finishing a session.
- **Dose-Response**: Among PTT users, longer average press duration was positively correlated with both cards generated and characters per turn, supporting the hypothesized mechanism that manual control allows more thoughtful, uninterrupted narration.

5.2 Practical Implications

The results provide moderate evidence in favor of PTT for enhancing content richness in memory journaling applications. While statistical significance was not achieved—likely due to the limited sample size—the consistent directional trends and the plausible mechanism (avoiding premature cut-offs) support a

product recommendation to offer PTT as an optional or default mode. Further testing with a larger, longer-duration experiment is recommended to confirm these effects.

5.3 Limitations

- **Sample Size**: With only 51 active users, power to detect small-to-medium effects was limited.
- **Short Duration**: The three-day window precluded analysis of retention or habituation.
- **Self-Selection**: Participants were recruited from the developers' network, which may not represent the broader target audience.
- **Missing Subjective Data**: The planned "Was your turn cut off?" question was not implemented, limiting insight into perceived interruptions.
- **Regional Imbalance**: The China deployment had only 2 users in Arm A, so subgroup analyses by region were underpowered.

6 Conclusion

This A/B test compared VAD and PTT voice interactions in the Yiliu app. While no statistically significant differences emerged in the primary outcomes, PTT users showed a consistent trend toward generating more memory cards and longer turns. The dose-response relationship further supports the hypothesis that manual turn control benefits narrative depth. We recommend a follow-up experiment with a larger sample to confirm these findings and inform the long-term product strategy.